Low Temperature Solder Interconnection of Surface Mount Devices with Additively Printed Pads on Flexible Substrate

Pradeep Lall
Auburn University
NSF-CAVE3 Electronics Research Center
Department of Mechanical Engineering
Auburn, AL 36849, USA
Email: lall@auburn.edu

Kartik Goyal
Auburn University
NSF-CAVE3 Electronics Research Center
Department of Mechanical Engineering
Auburn, AL 36849, USA

Kyle Schulze
Auburn University
NSF-CAVE3 Electronics Research Center
Department of Mechanical Engineering
Auburn, AL 36849, USA

Scott Miller
NextFlex National Manufacturing Institute
San Jose, CA 95131, USA

Abstract*— Printed electronics are rapidly increasing and competing against the traditional way of manufacturing electronics. Different aspects of the processes are already being taken over that avoids material wastage, reduces cost and lead times. For example, printing technologies are very popular in depositing reliable and very fine conductive traces as opposed to conventional etching. Previously, authors have developed process studies by utilizing Aerosol Jet Printing technology in realizing multi-layer metallization with successive metal and dielectric layers and additively printing a donut micro-via. However, the merits of printing technologies do not stop here. Recently, assembly of Surface Mount Devices (SMD) on a flexible substrate has been of interest that requires low processing temperatures. Conventionally, this is done by components mounted on a solder paste and reflowing to about 220 °C. This temperature is extremely high and induces many reliability concerns. With the use of printing technologies, this can be avoided. A viable method would be to utilize low temperature solder for which the peak reflow temperature could be much lower than 220 °C. Nevertheless, with new methods, there comes new complications. For example, reliability concerns involving the interconnection joints between SMD and substrate. Since this is to be done additively, another concern comes about the printer's capability to dispense the solder paste as well as the metal conductive pad simultaneously. Having to additively print conductive pad, and screen print the solder brings about much manual work and just not feasible. In this work, authors have utilized state-of-art micro-dispense additive printer to dispense the metal conductive ink and the Low Temperature Solder (LTS) for SMD attachment. Process studies are done that addresses the physical profile of the low temperature solder for various print parameters. Multiple SMDs such as resistors, capacitors, and Inductors are mounted with semi-automatic pick-and-place machine. The reflow will still be done but at a lower temperature. Process development of additive part of the assembly will be done, along with the characterization of physical, electrical and mechanical properties.

Keywords*— direct write, printed electronics, low temperature solder, flexible substrate, surface mount devices.

Nomenclature

SMD	Surface Mount Detectors
LTS	Low Temperature Solder
FHE	Flexible Hybrid Electronics
CAD	Computer Aided Design
cP	Centipoise, unit of viscosity
DW	Direct Write
LED	Light Emitting Diode
IPA	Isopropyl Alcohol
COTS	Commercial off the Shelf components
BGA	Ball Grid Array package

I. Introduction

Flexible Hybrid Electronics (FHE), a field that is continuously evolving and expanding its footprint, is already on its way to be the next revolution in printed electronics industry. A recent report from IDTechEx finds that the global demand for FHE will reach a value of over $3 billion by 2030 [1], i.e. a huge market potential. The term 'FHE' is loosely translated as the integration of additive printed conductive elements with traditional manufactured hardware. The reason for this integration is mainly because of the benefits that additive technologies provide that can be difficult to achieve with conventional processes. The major benefit comes with the flexible substrate, usually a polymer that can conform to any type of surface. This type of substrate comes out to be quite incompatible with the resists and etchants that are used in conventional lithography. Another benefit with additive processes involves the single-step process in which a single CAD file is needed for patterning as opposed to multi-step etching. These benefits and more combine together to provide an excellent alternative for the manufacturing of low-cost flexible electronics. Different additive technologies are being explored that can match the resolution achieved from conventional lithography, and also to be compatible with various materials. Some of them include Aerosol Jet, DirectWrite, and Inkjet. Each of these can achieve a resolution of <10μm. But with respect to materials, one technology cannot

incorporate all kinds of materials, such as pastes, inks. For example, Aerosol Jet is capable to material viscosity ranging from 1-1000cP, Direct Write is capable to dispensing thick viscous pastes up to 100,000cP, Inkjet limited to narrow 5-30cP materials. Whatever the limitation, each of these can offer something better than the other.

Much work has been published in the field of printed electronics. Although the production has been around for decades, the development is still ongoing that allows the manufacturers and researchers to bypass the conventional manufacturing techniques. A detailed review on the state-of-art technologies for different direct-write printing of electronics is discussed in [1], with materials, processes, and applications for flexible printed electronics discussed in [3]-[4]. Process studies are also investigated involving the conductive inks that are used, such as [5] discussing the fabrication of tracks using Inkjet printing, [6] characterizing the metal-based Inkjet inks, [7] focusing on the comparison of Microparticle and Nanoparticle based Ag inks. Demonstrations of the additive technologies have been shown in various forms including antennas [8]-[10], RFID tags [11], harmonic transceiver [12], and passive electronic elements [13]-[15]. Even though many electrical devices have been completely additively printed, such as thin-film transistors [16], inductors [17], capacitors [18], and batteries [19], it is not feasible to replace them all in an electronic system because of the yield and performance as compared to traditional silicon components. Hence, the rise of hybrid printed electronics, that involve combination of printed and COTS components. Therefore, it is imperative to study the interconnection materials that can be deposited between the printed tracks and the component. Conductive epoxies and adhesives have been exploited to achieve the interconnection using Inkjet technology [20]-[21], flip chip with conductive bumps on paper assembly technology [22]. Other than conductive epoxies, low temperature solders have been recently gaining interest to avoid local thermal damage to heat sensitive elements, but not much open literature can be found to address the resiliency of this material. Most commonly, low temperature solders consists of bismuth (Bi) or indium (In) alloys with tin (Sn) and small addition of silver (Ag). Bi tend to have higher strength but reduced ductility while, In lower strength and higher ductility. In also tends to be expensive than Bi. Some of the studies with low temperature solders consists of characterization of flip chip solder joints [23], thermocompression bonding in Ag-In solders for packaging applications [24], quantifying the properties of Sn-Ag-Bi solders [25].

Process development of additive interconnections with low temperature solders is an area where literature gap seems to be vast and needs to be explored more. Often times, with any material, the properties are listed or provided for the best-case scenario. In the article, this gap is addressed by process studies that greatly impact the material characteristics including physical and electrical properties. With these additive technologies, there are multiple parameters that greatly affect the quality of the print. In this paper, DW technology is used to realize additive interconnections. First, a metal conductive pad is printed with silver ink. Next, a low temperature solder is printed on top of the pad for the interconnection with SMD. After the SMD is mounted using semi-automatic pick-and-place machine, the substrate is subjected to reflow. Since this is a low temperature solder, the peak temperature is much lower than the standard lead-free solder.

A. Direct Write Micro-Dispensing Technology

Micro-dispensing or direct-write is another digital printing technique that does a remarkable job in additively printing wide range of materials. It does so in exceptionally small volumes with very high precision, that is achieved because of its ability to dispense much closer to the substrate surface. Micro-dispensing is loosely translated as dispensing of material using a certain amount of pressure and pressing it out through a small orifice onto a substrate. To get into more details about the process and the technology, the discussion will be focused on one of the products from nScrypt, Inc, who have become the pioneers in this technology. The work in this paper is done on a 3Dn Tabletop printer that consists of a quad head assembly with SmartPump^{TM} and nFD^{TM} gizmos. The system also comes with a process view camera and laser for alignment purposes, if necessary. Since the materials used are conductible metal pastes, and considering that the nFD^{TM} is filament fed material extrusion tool, more details on the SmartPump^{TM} gizmo and its working are discussed next.

The SmartPumpTM gizmo mainly consists of a valve rod, the most important component in the gizmo housing that controls the flow of material extrusion. It requires the material to be loaded into a syringe, with one end connected to the pressure tube and the other to the valve body. The “open” and “close” position of the valve rod defines when to let the material flow and when to stop it without any kind of leaking or drooling. This movement of valve rod happens automatically, once the positions are defined, in a coordinated manner as per the requirement of design file. These positions along with few other process parameters can have a drastic impact on the quality of the print. Some of them include print speed, material pressure, nozzle height, that need to be set as well. The benefit of not having a complicated setup with the material being well-contained in a syringe makes it a perfect additive tool for advancement of the manufacturing of 3D printed electronics. More details on this and other gizmos can be found on the company website [26].

Heated nFD Pumps and SmartPumps

nScrypt

Fig. 1. Direct Write Micro Dispensing Printer in-House

The DW printer used in this study is from nScrypt, 3Dn-TableTop with quad-head assembly consisting of two SmartPumps™ for material dispensing and two nFD™ gizmos for filament-based material extruder. The printer with close-up view of the gizmos is shown in Fig. 1. The figure shows the material loaded up in syringe connected to a pressure hose. The syringe is plugged into one of the SmartPump™ on which a ceramic nozzle tip is attached.

II. Experimental

One of the advantages of the additive printing technology is the ability to deposit any type of pattern, if provided with a simple CAD file, keeping the resolution of printer in mind. The test design for this paper was modeled in such a way that the interconnection between SMD device and flexible substrate is achieved. The modeling was done in AutoCAD software for the generation of appropriate DXF file needed for DW printing. In the design, common SMD discrete such as resistors, inductors are included. Additionally, a LED is also mounted for verification of electrical property. The design is shown in Fig. 2.

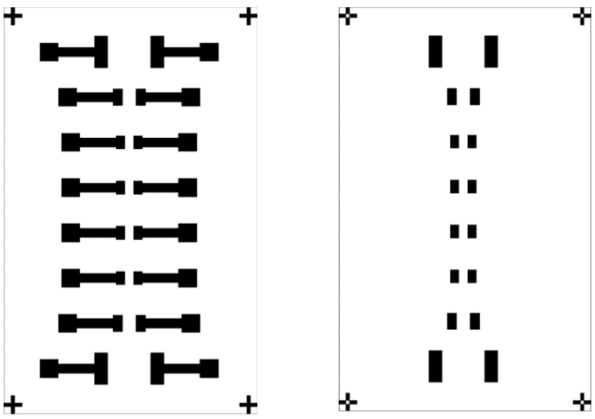

Fig. 2. Test Design for SMD Interconnection

A. Material Properties

The DW printing technology used in the paper utilizes a thick viscous paste as its depositing material. Therefore, a silver metal conductive paste is used to deposit the trace with SMD footprint and a low temperature solder paste is used to attach the SMDs. Low temperature solder was selected because of its low temperature processing and the solderable ability of the silver paste. The low temperature solder is also additively printed that allows the whole manufacturing process to be additive. Typical properties of both the materials are listed in TABLE I and TABLE II.

TABLE I. Typical Properties of Silver Nanoparticle Ink

Typical properties of Silver ink	
Material	Silver
Ag Content (wt. %)	70
Particle Size (μ)	1.5 Flake-based
Viscosity (cP)	105,000 at 0.1 sec^{-1} 22,000 at 100 sec^{-1}
Curing Conditions	175°C for 5 mins

TABLE II. Typical Properties of Low Temperature Solder

Typical properties of Low Temperature Solder	
Physical State	Paste, Pb-free
Composition	57Bi/42Sn/1Ag
Viscosity (Kcps)	1,600
Metal Loading (%)	80-90
Curing	Reflow (liquidus 150°C)

B. Procedure

Considering the complete process is to be additive, the process control of achieving the interconnection is imperative. Initially, a silver metal conductive paste is used to deposit the trace and pad geometry. The paste is thermally sintered to evaporate any surfactant using the recommended conditions. The design is brought over again, and after aligning, the low temperature solder is deposited on the SMD footprints. The components are mounted using a semi-automatic pick-and-place instrument. After the mounting of SMDs, the assembly was subjected to a reflow profile, shown in Fig. 3. The flowchart depicting the process is shown in Fig. 4.

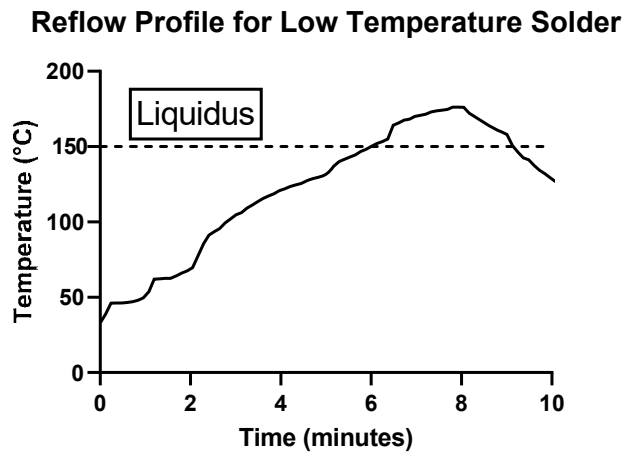

Fig. 3. Reflow Profile Used

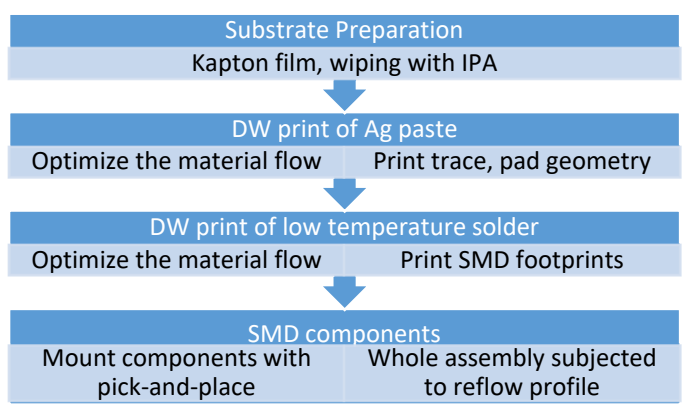

Fig. 4. Flowchart of Printing Process

III. Results and Discussion

Results of the process study done on DW printing is discussed in the following sections. The section is divided into

two sub-sections: DW print of silver and low temperature solder, and mounting of SMD components.

A. DW Printing

Direct write or micro dispensing technology is used for depositing silver tracks and low temperature solder material. As discussed in Introduction section, with direct write, the open and close valve rod position is the most important parameter that controls the flow of material. Other parameters such as pressure, print speed, and layer height are also influential to some extent. The effect of these parameters need to be studied in detail to optimize the process flow for SMD interconnection. The process parameters for dispensing of the silver metal paste will not be discussed in this paper, however the surface profile for the printed material is shown. For the low temperature solder, the discussion is in the following section.

Silver Metal Ink

Silver metal ink is used for printing of the traces and SMD pads using DW. The process of optimizing the print parameters are discussed in another paper. The surface profile is shown in Fig. 5 for the finalized parameters listed in TABLE III.

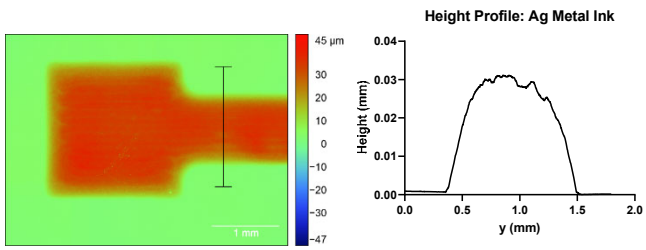

Fig. 5. 3D Surface Profile of Conductive Track

TABLE III. Finalized Printing Parameters for Metal Ink

Finalized Parameters for Silver Metal ink	
Print Speed (mm/s)	3
Material Pressure (psi)	18
Layer Height (mm)	0.050
Nozzle Diameter (mm)	0.159

Low Temperature Solder

Different process parameters are varied to see the effect on the surface profile of the solder material, such as print speed, layer height, and material pressure. Since the substrate, polyimide, is non-wettable, solder material after proper reflow does not stick to it and the effect of process parameters cannot be studied. Hence, the surface profile of solder is looked before the reflow initially. For after the reflow, the interface is studied for three different scenarios: solder and polyimide; solder and silver; and solder and copper film.

Low temperature solder is additively dispensed on polyimide film. Before printing, polyimide is manually wiped with IPA. Different process parameters are varied to study the print consistency via surface profile. Fig. 6 shows the effect of varying print speed, Fig. 7 shows the effect of varying layer height, and Fig. 8 shows the effect of varying material pressure.

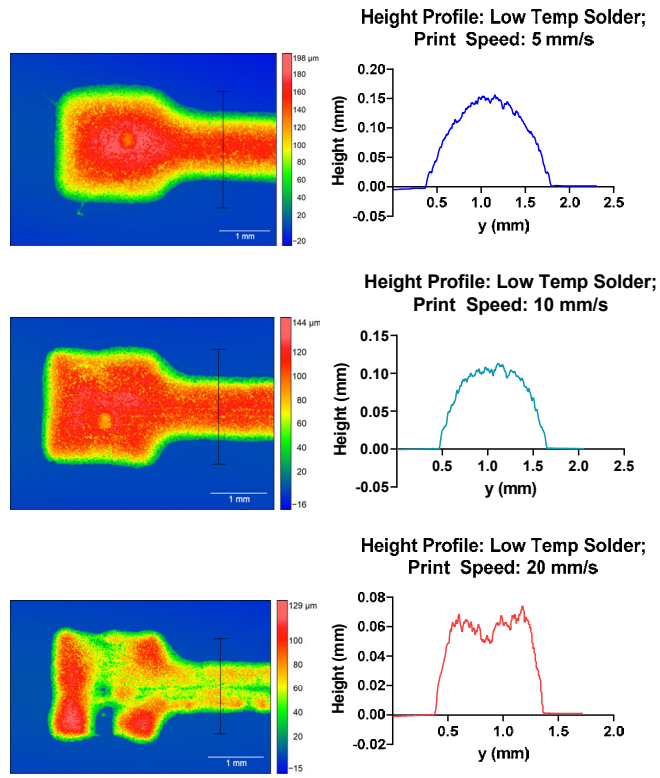

Fig. 6. Effect of Print Speed on Solder Profile

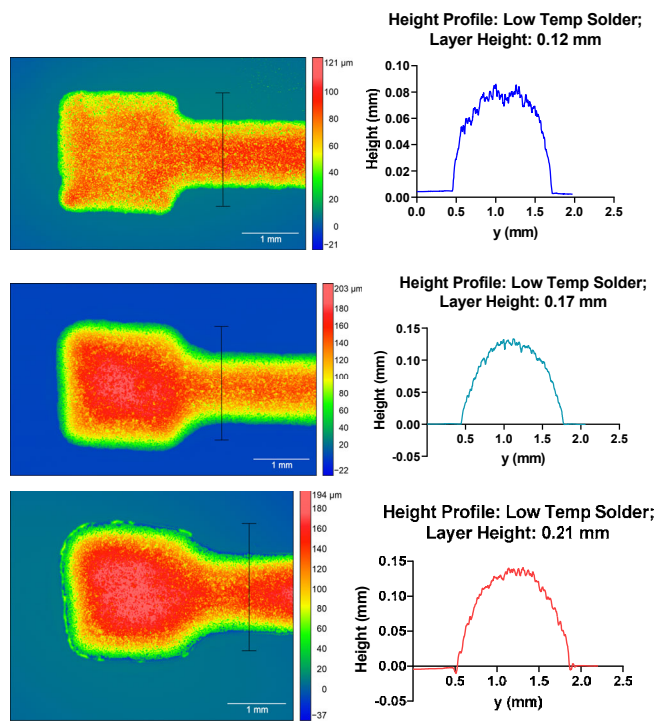

Fig. 7. Effect of Layer Height on Solder Profile

For the print speed, it is clear that the print, although shows lower height, starts to be inconsistent with speeds higher than 20 m/s. With layer height, the height of the print increases as it is expected, but there is negligible difference going above than

0.17µm. As a note, the layer height is the distance of nozzle tip from the substrate. Effect of pressure appears to be very significant. The pressure on the lower and higher end, considering the viscosity of the solder, the material behaves erratically with having an optimal range around 15 psi.

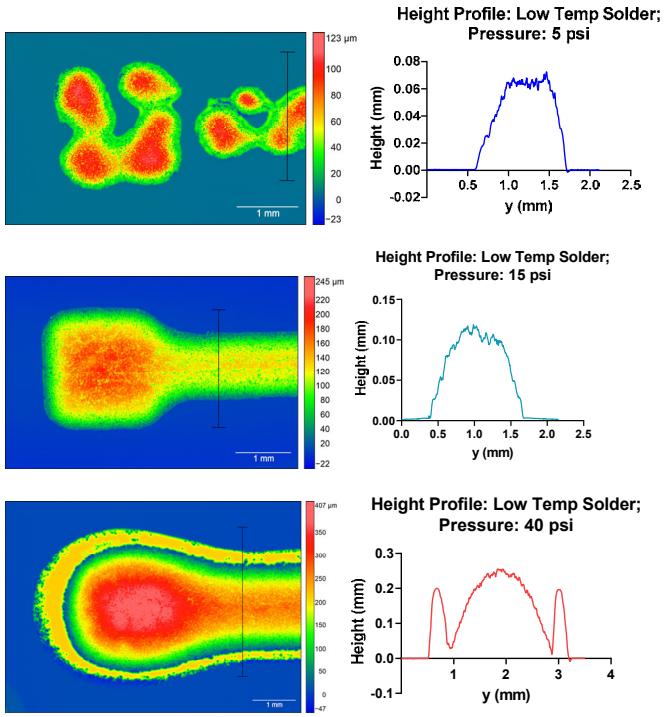

Fig. 8. Effect of Material Pressure on Solder Profile

Looking at the surface profiles for various parameters, a list is defined for the finalized parameters, shown in TABLE IV, and the surface profile is studied. The profile is shown in Fig. 9.

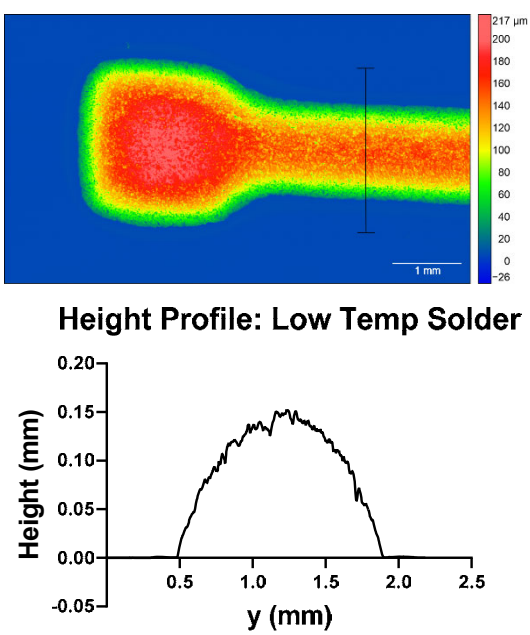

Fig. 9. Solder Profile for Finalized Printing Parameters

TABLE IV. Finalized Parameters for Solder Dispensing

Finalized Parameters	
Print Speed (mm/s)	10
Material Pressure (psi)	18
Layer Height (mm)	0.17
Nozzle Diameter (mm)	0.234

For the interface study, the solder material was printed on three different materials: polyimide film, silver, and copper film. Since polyimide is a non-wettable substrate, solder material after the reflow does not stick to the substrate. However, for silver and copper, it was important to study the interaction and see if there is any intermetallics formation between the two materials. Fig. 10 shows the solder material on polyimide after the reflow. The modeled CAD file consisted of 1 mm circles. After reflow, the material reduced to a ball, leaving the flux around it. The bulk solder material is comprised of Bi-rich phase (dark contrast) and Sn-rich phase (light contrast).

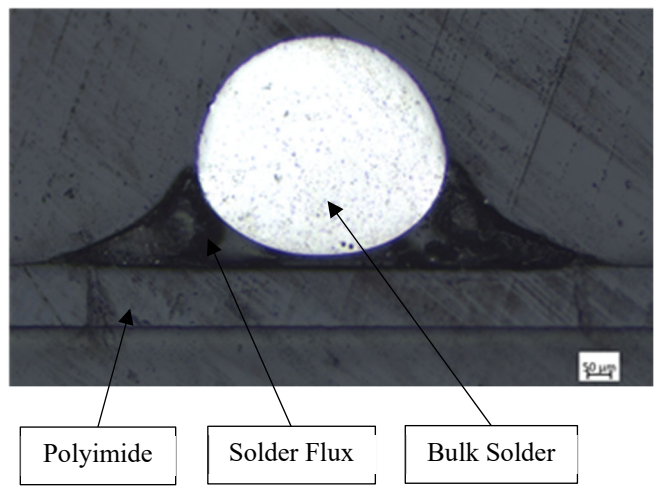

Fig. 10. Solder on Polyimide Film

Solder behaves differently when dispensed on a wettable material, such as silver or copper. Fig. 11 shows an optical picture of solder on silver. The interface does not show any intermetallics formation on the pristine sample. For the solder on copper film, shown in Fig. 12, there is possible copper-tin intermetallics formation.

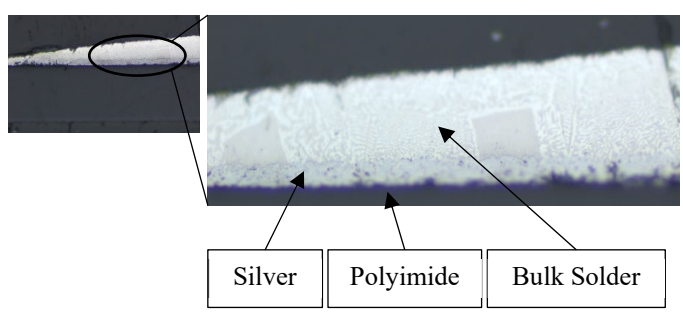

Fig. 11. Solder on Silver Track on Polyimide

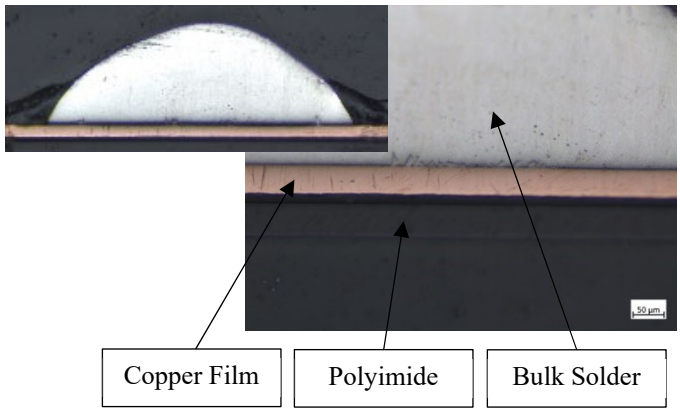

Fig. 12. Solder on Copper Film

Electrical resistivity of the additively deposited low temperature solder was also quantified and compared with the properties published in literature. To measure the resistivity, silver metal line is printed as a base layer, on top of which solder was deposited. Measured resistance was subtracted from the resistance of metal base layer, which was used for resistivity calculation. Fig. 13 shows the variation in electrical resistivity of the solder for about eight samples. The average resistivity of 57Bi-42Sn-1Ag was found to be about 3 times the resistivity of 58Bi-42Sn. Resistivity of 58Bi-42Sn was obtained from published literature [27].

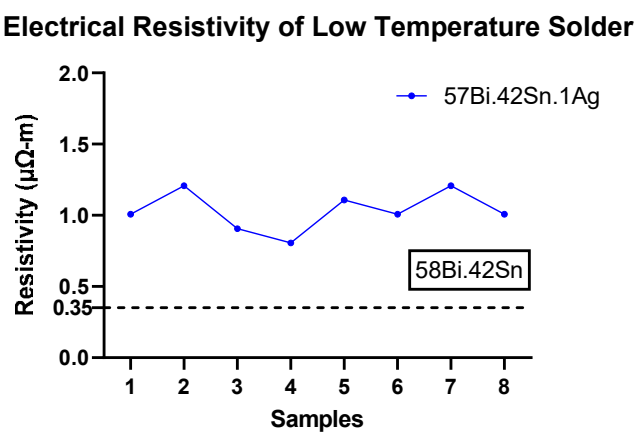

Fig. 13. Electrical Resistivity of Low Temperature Solder

B. *SMD Discrete Attachment*

Surface mount discretes are mounted via complete additive process. A solderable silver ink is dispensed initially for the SMD footprint and trace pad, followed by low temperature solder on the SMD footprint. The discretes are mounted using a semi-automatic pick-and-place machine, and the whole assembly is put through reflow. With the low temperature solder, the peak reflow temperature is much lower (160-165°C) as compared to standard lead free SAC solders (240-250°C), that can be beneficial if the components cannot sustain high temperatures. Both the materials are dispensed using the DW printing technology. Fig. 14 (left) shows the SMDs resistors and inductors mounted on the flexible substrate. A LED is also mounted for display of electrical connectivity as well, shown in Fig. 14 (right).

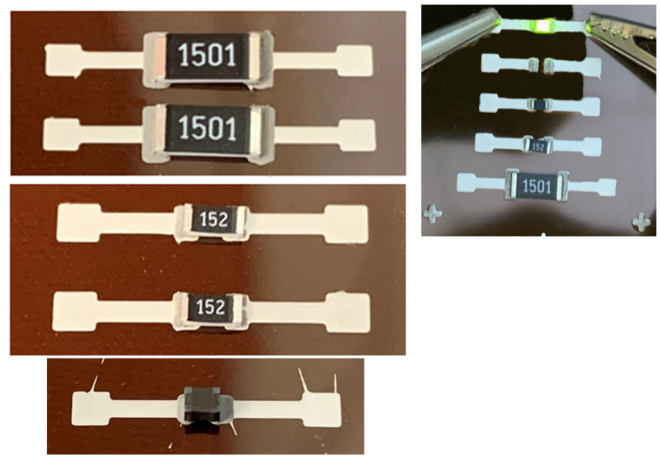

Fig. 14. SMDs on Direct Write Deposited Tracks and Low Temperature Solder

Repeatability in mounted SMD resistance is also quantified. Sample to sample variation was found to be negligible relative to the rated resistance of 1.5 kΩ, as shown in Fig. 15.

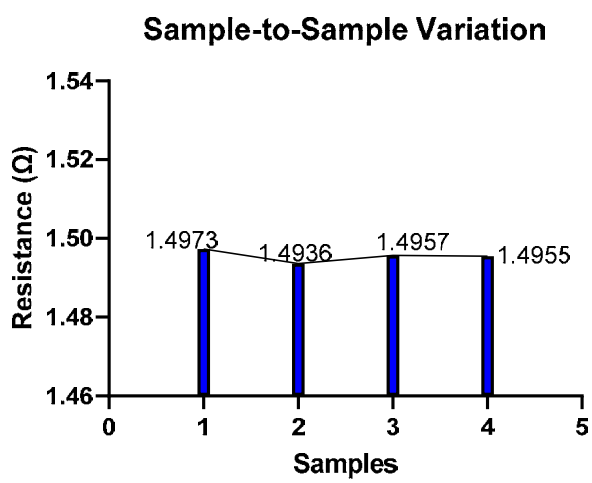

Fig. 15. Sample to Sample Variation of SMD Resistor

Multiple types of SMD discrete components are attached including resistors, capacitors, and inductors. The property of each of the component was measured at various frequency levels to look at the variation. Fig. 17-Fig. 19 shows the effect of frequency of resistance, inductance, and capacitance respectively, relative to the rated values. These measurements are done using an Agilent 4192A LF Impedance Analyzer. Not much variation is seen for inductor and capacitor at various levels of frequency, however, for resistor, there is a huge drop in resistance at 500 kHz. This drop is due to the parasitic capacitance effect that SMD resistor exhibits at higher frequency levels. At lower frequency levels, this effect is not as dominant and resistor acts as an ideal resistor.

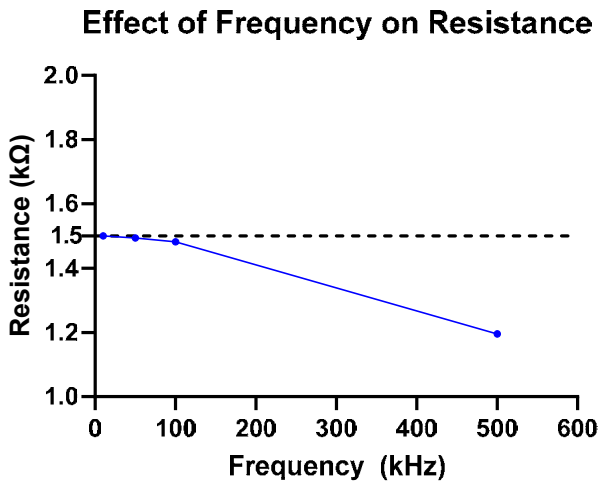

Fig. 16. Frequency Variation in Additively Mounted SMD Resistor

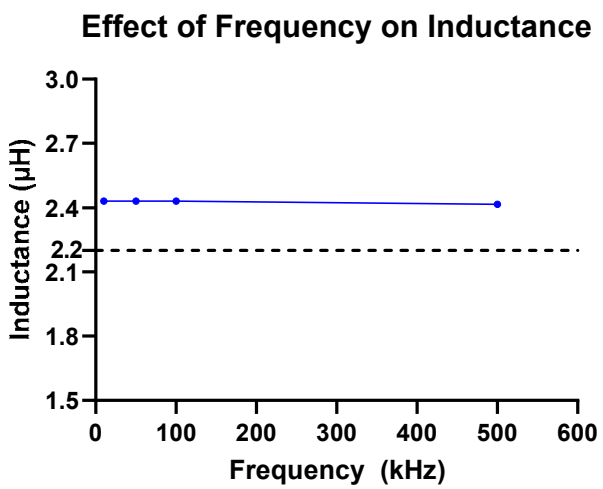

Fig. 17. Frequency Variation in Additively Mounted SMD Inductor

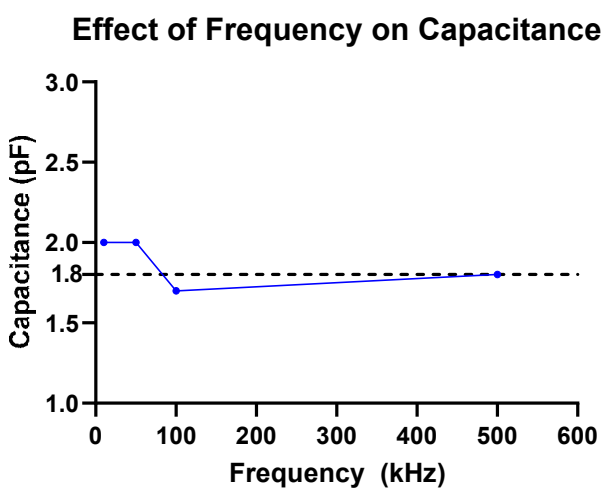

Fig. 18. Frequency Variation in Additively Mounted SMD Capacitor

A daisy-chain BGA is also mounted to show the potential of DW technology and the capability of process developed for SMD attachment. The BGA selected is PBGA 256, as shown in Fig. 19. Optical pictures of the cross section of mounted SMD and BGA solder joint are also shown in Fig. 20. For the solder joint, the optical pictures shows the failed solder joint. This failure occurred due to the bending of the polyimide film.

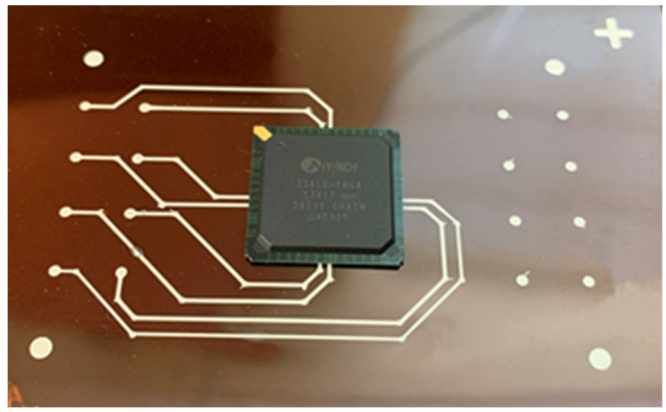

Fig. 19. A daisy-chain Sample via Complete Additive Direct-Write Process

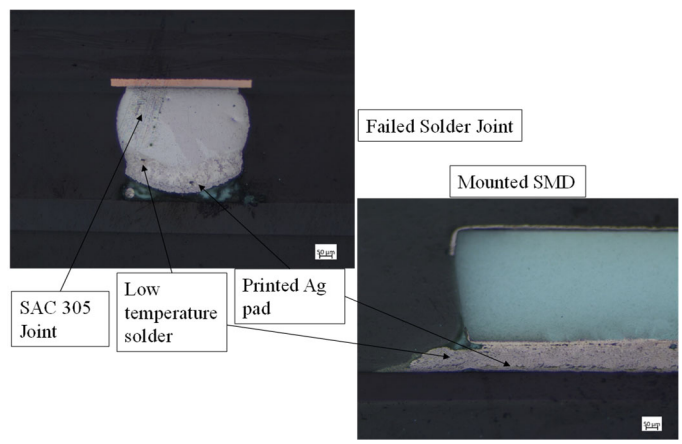

Fig. 20. Cross-sectional Pictures of BGA Solder Joint and SMD Package

IV. Summary and Conclusions

Direct write micro-dispensing technology is one of the few technologies that continue to prove its efficiency with fine print quality and features. Moreover, its ability to dispense wide range of materials makes it a very useful gadget for printed electronics. However, it does require control of few parameters. The most important comes to be the timing of material flow, specifically, open and closed position of valve rod that dictates when to allow the material to flow and when to stop. Since this valve rod position cannot be varied, other parameters that can have a significant impact on the print quality are studied. Some of them include material pressure, layer height, and print speed.

To achieve additive interconnection between a SMD and flexible substrate, two materials are used: a solderable metal conductive ink, and low temperature solder, for which the peak reflow temperature was much lower (160°C) than standard SAC solders. Process parameters were only varied for the low temperature solder, and effect of those on the surface profile was presented. A point to note is that the surface profiles were

studied before the solder was subjected to reflow to look at the printed features. Since the print is done on polyimide substrate, which is extremely non-wettable and hydrophobic, the solder cross section is studied after the reflow. The solder on polyimide does not wet the substrate and after the reflow turns into bulk solder with flux around it. Other than polyimide, solder is also dispensed on silver, and copper film as well. With silver and copper film on the bottom, solder wets the surface. Although for copper, not as much. There is intermetallics formation as well. To show the capability of direct write micro dispensing printing technique, finally, SMDs are mounted. A green LED is also mounted for the verification of electrical connectivity. Additionally, a daisy-chain BGA component was also mounted on the flexible polyimide to show the potential of additive technology. Quantification of some of the process parameters has been presented to dispense a novel low temperature solder material for additive interconnections between SMD and the substrate. Also, to show the capability of DW technology by mounting SMDs, including a BGA package.

ACKNOWLEDGEMENTS

The project was sponsored by the NextFlex Manufacturing Institute under PC4.5 Project titled – Multilayer Printed Transmit Antenna with Integrated Electronics. This material is based, in part, on research sponsored by Air Force Research Laboratory under agreement number FA8650-15-2-5401, as conducted through the flexible hybrid electronics manufacturing innovation institute, NextFlex. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright notation thereon. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of Air Force Research Laboratory or the U.S. Government. Authors would also like to acknowledge Emily Guo of Auburn University for helping with White Light Interferometry scans of printed conductive line sample.

REFERENCES

[1] Dyson, Matthew, et al. "3D Electronics 2020-2030: Technologies, Forecasts, Players." IDTechEx, 12 Aug. 2020, www.idtechex.com/en/research-report/3d-electronics-2020-2030-technologies-forecasts-players/769.

[2] Zhang, Y., Liu, C., & Whalley, D. (2009, August). Direct-write techniques for maskless production of microelectronics: A review of current state-of-the-art technologies. In *2009 International Conference on Electronic Packaging Technology & High Density Packaging* (pp. 497-503). IEEE.

[3] Huang, Q., & Zhu, Y. (2019). Printing conductive nanomaterials for flexible and stretchable electronics: A review of materials, processes, and applications. *Advanced Materials Technologies*, *4*(5), 1800546

[4] Cui, Z. (2016). *Printed electronics: Materials, technologies and applications*. John Wiley & Sons.

[5] Chen, S. P., Chiu, H. L., Wang, P. H., & Liao, Y. C. (2015). Inkjet printed conductive tracks for printed electronics. *ECS Journal of Solid State Science and Technology*, *4*(4), P3026

[6] Kamyshny, A., Steinke, J., & Magdassi, S. (2011). Metal-based inkjet inks for printed electronics. *The Open Applied Physics Journal*, *4*(1).

[7] Roberson, D. A., Wicker, R. B., Murr, L. E., Church, K., & MacDonald, E. (2011). Microstructural and process characterization of conductive traces printed from Ag particulate inks. *Materials*, *4*(6), 963-979.

[8] Arnal, N., Ketterl, T., Vega, Y., Stratton, J., Perkowski, C., Deffenbaugh, P., ... & Weller, T. (2015, May). 3D multi-layer additive manufacturing of a 2.45 GHz RF front end. In *2015 IEEE MTT-s International Microwave Symposium* (pp. 1-4). IEEE.

[9] Church, K. H., Crane, N. B., Deffenbaugh, P. I., Ketterl, T. P., Neff, C. G., Nesbitt, P. B., ... & Wang, J. (2017). Multimaterial and multilayer direct digital manufacturing of 3-D structural microwave electronics. *Proceedings of the IEEE*, *105*(4), 688-701.

[10] Ketterl, T. P., Vega, Y., Arnal, N. C., Stratton, J. W., Rojas-Nastrucci, E. A., Córdoba-Erazo, M. F., ... & Weller, T. M. (2015). A 2.45 GHz phased array antenna unit cell fabricated using 3-D multi-layer direct digital manufacturing. *IEEE Transactions on Microwave Theory and Techniques*, *63*(12), 4382-4394.

[11] Ramirez, R. A., Rojas-Nastrucci, E. A., & Weller, T. M. (2017). UHF RFID tags for On-/Off-metal applications fabricated using additive manufacturing. *IEEE Antennas and Wireless Propagation Letters*, *16*, 1635-1638.

[12] Nassar, I., Tsang, H., & Weller, T. (2014). 3D printed wideband harmonic transceiver for embedded passive wireless monitoring. *Electronics Letters*, *50*(22), 1609-1611.

[13] Ramirez, R. A., Rojas-Nastrucci, E. A., & Weller, T. M. (2018). Laser-assisted additive manufacturing of mm-wave lumped passive elements. *IEEE Transactions on Microwave Theory and Techniques*, *66*(12), 5462-5471.

[14] Koshi, T., Nomura, K. I., & Yoshida, M. (2020). Electronic Component Mounting for Durable E-Textiles: Direct Soldering of Components onto Textile-Based Deeply Permeated Conductive Patterns. *Micromachines*, *11*(2), 209.

[15] Guan, X., Cao, L., Huang, Q., Kong, D., Zhang, P., Lin, H., ... & Yuan, H. (2019). Direct Writing Supercapacitors Using a Carbon Nanotube/Ag Nanoparticle-Based Ink on Cellulose Acetate Membrane Paper. *Polymers*, *11*(6), 973.

[16] Sowade, E., Ramon, E., Mitra, K. Y., Martínez-Domingo, C., Pedró, M., Pallarès, J., ... & Baumann, R. R. (2016). All-inkjet-printed thin-film transistors: manufacturing process reliability by root cause analysis. *Scientific reports*, *6*, 33490.

[17] Gu, Y., Park, D., Gonya, S., Jendrisak, J., Das, S., & Hines, D. R. (2019). Direct-write printed broadband inductors. *Additive Manufacturing*, *30*, 100843.

[18] Correia, V., Mitra, K. Y., Castro, H., Rocha, J. G., Sowade, E., Baumann, R. R., & Lanceros-Mendez, S. (2018). Design and fabrication of multilayer inkjet-printed passive components for printed electronics circuit development. *Journal of Manufacturing Processes*, *31*, 364-371.

[19] Pang, Y., Cao, Y., Chu, Y., Liu, M., Snyder, K., MacKenzie, D., & Cao, C. (2020). Additive manufacturing of batteries. *Advanced Functional Materials*, *30*(1), 1906244.

[20] Andersson, H. A., Manuilskiy, A., Haller, S., Hummelgård, M., Siden, J., Hummelgård, C., ... & Nilsson, H. E. (2014). Assembling surface mounted components on ink-jet printed double sided paper circuit board. *Nanotechnology*, *25*(9), 094002.

[21] Arrese, J., Vescio, G., Xuriguera, E., Medina-Rodriguez, B., Cornet, A., & Cirera, A. (2017). Flexible hybrid circuit fully inkjet-printed: Surface mount devices assembled by silver nanoparticles-based inkjet ink. *Journal of Applied Physics*, *121*(10), 104904.

[22] Rasul, J. S. (2004). Chip on paper technology utilizing anisotropically conductive adhesive for smart label applications. *Microelectronics Reliability*, *44*(1), 135-140.

[23] Kang, U. B., & Kim, Y. H. (2004, June). Electrical characteristics of fine pitch flip chip solder joints fabricated using low temperature solders. In *2004 Proceedings. 54th Electronic Components and Technology Conference (IEEE Cat. No. 04CH37546)* (Vol. 2, pp. 1952-1958). IEEE.

[24] Made, R. I., Gan, C. L., Yan, L. L., Yu, A., Yoon, S. W., Lau, J. H., & Lee, C. (2009). Study of low-temperature thermocompression bonding in

Ag-In solder for packaging applications. *Journal of electronic materials*, *38*(2), 365.

[25] Kim, K. S., Imanishi, T., Suganuma, K., Ueshima, M., & Kato, R. (2007). Properties of low temperature Sn–Ag–Bi–In solder systems. *Microelectronics Reliability*, *47*(7), 1113-1119.

[26] *nScrypt – Precision Microdispensing and Direct Digital Manufacturing*. (n.d.). NScrypt. https://www.nscrypt.com.

[27] Siewert, T., Liu, S., Smith, D. R., & Madeni, J. C. (2002). Properties of lead-free solders. *Database for solder properties with emphasis on new lead-free solders, Release*, *4*